D · Teacher guidance, alignment and marking

WS2 v4 • VL2-Q3 • VL2-R3

v4 keeps four pages and uses the full native VL anatomy and cutaway artwork. Orange 1a/1b, purple 2a/2b and blue 3 match the three note paragraphs. Variables, assumptions and control are items 1, 2 and 3. The final question is taken verbatim from the VL source, including existing English vocabulary support.

Distribution and shared time

The control group receives no prior concept lecture and does not operate the VL. Distribute the complete student worksheet at the start, with topic notes available throughout. Collect or scan first responses before distributing complete answers. Keep original responses and correct in another colour. D is for teachers only.

Shared target: 15–20 minutes: notes 3–4, context and fair comparison 4–5, measurement/calculation/graph 5–6, analysis/integration 3–4, saving/checking about 1 minute. Pilot the workload; completion by every pupil is not guaranteed. If checking takes longer, apply the same arrangement to both groups and record it.

Clarify procedure, choices, scale position and units only. Before saving first responses, give no concept teaching or correct choices. Record support, unfinished items and time. VL learning points and reflection remain locked until investigation submission; student PDF unlocks after reflection submission.

Item alignment, format and shared content

WS	VL	Format / shared concept or skill
1	variableQuiz	Seven full choices per row: IV/DV/CV
2	assumptionChoices	Original four choices: fair-comparison assumptions
3	controlPlan	Open design: equal-distance leaf comparison
4	measurements 0/10/20/30	16 scale readings: observation and recording
5	rate-A/B/C/D	Four calculations: own initial/final readings
6	graph.points / connected	Four points and A/B/C trend: communication
7	claim	Original sentence choice: light trend

WS	VL	Format / shared concept or skill
8	leafComparison	Original sentence choice: B/D comparison
9	leafConclusion	Original sentence choice: leaf function
10	limitations	Original three choices: indirect indicator
11	reflection (Q3)	Integrated explanation: transpiration and pull

Paper has no response opportunity for original preliminary observation, pre-experiment prediction/reason, free apparatus drawing/written plan, ruler dragging, simulation timing or confirmation actions. Static scale reading is not a full practical investigation. Concept-first study changes prediction opportunities, so no hypothesis-revision question is invented.

Paper item marking and omissions

Record item scores separately; do not convert a paper activity total to the VL 32-point scale. Mark an omitted asked item 0, an unmarked item blank, and an unasked item not applicable. Judge concepts and causes; accept equivalent wording rather than counting keywords.

Item	Max	Credit and partial credit
1	4	IV exact 1; DV exact 1; CV 0.4 each; including IV/DV in CV gives CV 0.
2	1	Select only choices 1, 2 and 4: 1; otherwise 0.
3	1	1 for equal lamp distance, leaves versus no leaves, and other conditions equal.
4	2	2 × correct/16; tolerance ±0.25 cm; two decimals.
5	2	0.5 each; own (final−initial)/30; ±0.0051 cm/min.
6	2	0.375 per point against own calculations; correct A/B/C trend 0.5, D separate.
7–10	2	0.5 each for 7, 8, 9, 10; judge 8 against own B/D rates.
11	8	Use the Q3 mechanism rubric below; no hypothesis revision required.

Full answers and acceptable equivalents

3: Compare setup D without leaves with setup B with leaves. Both have a lamp distance of 20 cm and the same other conditions. The main difference is whether leaves are present.

1: IV light intensity; DV height/average upward speed of red trace; CV the remaining five factors. 2: choices 1, 2, 4.

Time min	A	B	C	D
0	0.0	0.0	0.0	0.0
10	4.0	2.5	1.5	0.5
20	8.0	5.0	3.0	1.0
30	12.0	7.5	4.5	1.5

5: A 0.40, B 0.25, C 0.15, D 0.05 cm/min. Accept correct calculation from own readings; do not penalise the same scale error again in calculation or plotting.

6: A(1200,0.40), B(300,0.25), C(133,0.15), D(300,0.05), with D separate. 7 increase; 8 faster (follow own data); 9 help water move upwards; 10 first choice.

11: With other conditions equal and an adequate water supply, stronger light usually favours **stomatal opening**. Water **evaporates** from the **water films** on mesophyll cell surfaces into the **air spaces**. Water vapour then **diffuses** through **stomata** down a concentration gradient and leaves the leaf, producing **transpiration**. Water loss lowers the **water potential** of mesophyll cells, and water from **xylem vessels** replaces it. Removal of water from the vessels creates **transpiration pull**, drawing water upwards through **xylem vessels**. Leaves provide the main surface for **transpiration**; a setup with leaves usually shows faster upward movement of the water trace than a leafless setup at the same lamp distance. Water-trace speed is an indirect indicator and is not the **transpiration** rate.

VL plotting retains exact assigned lux and Y within 0.00051 of own calculated speed. Judge paper points by their legible plotted position and own calculations; handwritten position is not digital input precision and requires piloting.

Q3 final item: eight-point rubric

Domain	Two independent points (1 each)
Transpiration / stomata	Water films evaporate into air spaces; vapour diffuses down a concentration gradient through stomata.
Xylem / pull	Loss lowers water potential and xylem replenishes water; removal creates pull and upward transport.
Light intensity	Other conditions equal, stronger light favours stomatal opening; connect to faster transpiration/water transport.
Leaves / comparison	Faster trace with leaves at equal distance; leaves provide the main transpiring surface and help draw water.

Each domain: 0/1/2. Do not require group-by-group citations, limitations or hypothesis revision. A contradictory cause earns no credit for that point. Accept equivalents such as diffusion from higher to lower vapour concentration.

Excel, historical versions and research use

Current export retains SPS24 + reflection8 = 32 for within-VL feedback. Q3 uses the leaves field; the old revision field is not applicable. Q1–2 retain their old rubric and have no leaves field. Q3 measurement-record quality is manual 0–2, rather than automatically treating completion as ability; historical round scores are retained. Other manual fields cover observation, hypothesis/reason, control, setup, communication, transpiration, xylem, light and the applicable final domain.

Manual inputs use whole-number validation from 0 to the maximum. Formula guards also reject pasted negatives, out-of-range values, fractions and text. Blank means pending; 0 means marked zero; not applicable does not block totals. Totals require valid applicable manual marks and a submitted reflection; colours follow valid scores.

Q3 record quality: traceable setup/time labels, units and records 1; water-level zero and consistent scale method 1. Judge records and plan; leave pending if evidence is insufficient. A reasonable original prediction can earn full credit even if unsupported. Events, clicks, speed, completion and time are not direct SPS scores.

Original/first/final answers, snapshots, setup images, events, research IDs and cloud queues remain intact. Student PDFs exclude teacher scores. Previously downloaded Excel files do not update automatically; a new download does not merge old manual marks. Common tests and marker agreement require separate validation; WS and VL do not offer identical response opportunities or an established common scale.

Curriculum, scientific limits and version

Checked against learning points and the existing repository citation to textbook section 11.1, pp.11-3–11-8. The original textbook and official syllabus were not supplied; official requirements are not independently verified. Objectives: fair-comparison variables; scale reading, calculation and plotting; indirect evidence versus mechanism; evaporation, diffusion, osmosis, xylem transport and transpiration pull.

Lamp distance can affect temperature; monitor it in a real experiment; the model fixes 25°C. Trace speed is indirect. Leaf removal alone cannot prove a stomatal mechanism, and three light levels do not justify an exact extrapolated curve. Xylem water is pulled upwards, not actively pumped or transported upwards in phloem.

This edition replaces A–C booklets and the combined teacher pack. Five download types per language: student PDF/Word, answers PDF/Word, D PDF. worksheet.py generates documents; source.js reads the actual VL model, choices and hints.